

SHENG ZHAO

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My research interest focus on designing efficient and trustworthy interaction systems between **human** and **XR** to enhance human well-being.

EDUCATION

Tsinghua University, Beijing, China

2021.09 – Present

B.Sc. in **Fundamental Science (Physics)** & B.Eng. in **Electrical Engineering**

Dual degree

- GPA: 3.81/4; Major GPA: 3.86/4.
- A+: Optics, Quantum Mechanics; A: Electromagnetism, Thermodynamics, Calculus, Mathematical Physics Equations, etc.

PUBLICATIONS & MANUSCRIPTS

- **Sheng Zhao**, JunRay Zhu, Xueyang Wang, Hongyi Li, Fang Yi, Hewu Li, Shuning Zhang and Xin Yi[†]. CoordAuth: A Two-factor Authentication Method in Virtual Reality Leveraging Head-Eye Coordination. *Rejected by IMWUT'24-a, in submission to CHI'25*. [PDF](#).
- Xueyang Wang*, **Sheng Zhao***, Yihe Wang, Ziyu Han, Xin Tong and Xin Yi[†]. EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in Virtual Reality. *TVCG, under review*.
 - * indicates equal contribution. [†] denotes the corresponding author.

RESEARCH EXPERIENCES

EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in VR

2023.04 – 2024.01

Research Assistant, Advisor: *Xin Yi, Yuanchun Shi*

Tsinghua University

My contributions:

- Implementing VR face2face chat using TCP-based socket and Oculus-Movement sdk in Unity.
- Designing both qualitative and quantitative research methods based on interviews and questionnaires, conducting user studies.
- Proposing an improved camera2blendshape method to mitigate the uncanny valley effect and enhance the emotional experience.

Submitted to *TVCG*. Under review. [Details](#).

CoordAuth: A Two-factor Authentication Method in VR Leveraging Head-Eye Coordination

2023.10 – 2024.01

Research Assistant, Advisor: *Xin Yi*

Tsinghua University

My contributions:

- Project leader, proposing the current two-factor authentication system based on unlock pattern and behavior biometrics.
- Using Unity to build the unlock UI leveraging gaze interaction in VR.
- Proposing a voting mechanism using two types of classifiers implemented by random forest to achieve behavioral biometric authentication factor and implementing the algorithm by python.
- Conducting user studies and the paper work.

Rejected by *IMWUT'24-a*. In submission to *CHI'25*. [PDF](#). [Details](#).

AvatarMVFusion: Text-Guided Multi-View Diffusion for Avatar Generation

2024.11 – Present

Research Assistant, Advisor: *Bowen Zhou*

Tsinghua University

My contributions:

- Fine-tuning diffusion model by dreambooth for avatar generation with high quality. [Details](#).
- Testing the performance of Zero123 (SOTA) and DreamFusion, figuring out the importance of prompt-driven multi-view generation for the avatar leveraging diffusion.
- Proposing a novel text-guided denoising 3D Unet framework and conducting experiments to evaluate its performance.

Preparing for *AAAI'24*.

Multi-avatars better? Investigating the Impact of Multiple Avatar Agents on Emotion Support in VR

2024.01 – Present

Research Assistant, Advisor: *Yukang Yan*

University of Rochester

My contributions:

- Project leader, proposing all the ideas and methods.
- Fine-tuning Llama3-8b on the ExTES dataset using QLoRA for base model.
- Building the multi avatar agents system, which can recognize user's speech and provide multimodal feedback.
- Designing the emotion support strategy and the coordination way of the agents.

Preparing for *CSCW'25-a*.

SKILLS

- Program languages: C#, Python, Java, MATLAB and Verilog.
- Tools and Frameworks: Git, LaTeX, Markdown, MSP430, Unity 3D, PyTorch, Scikit-learn, Huggingface, Transformers and Diffusers.

- Techniques: VR Skeleton Animation, ARKit, Quantitative and Qualitative research.

COMPETITION AWARDS AND HONORS

- *First Prize*, 36th China Physics Olympiad (CPHO), 0.9% 2020
- *First Prize*, The 12th Tsinghua University Digital System Design Competition, 1.7% 2022
Designed a small car driven by MSP430 with various sensors. [Details](#).
- *Second Prize*, Tsinghua University Software Design Competition 2023
Developed a Smart Home app using Java. [Code](#).
- *First Prize*, The 3rd Tsinghua Craftsman Competition, 1%, presented a speech 2023
- *First Prize*, Tsinghua University Technological Innovation Scholarship 2023
- *First Prize*, Tsinghua University Sports Excellence Scholarship 2022, 2023